

Statement

Machine learning based classification of credit card approval.

Dataset

From UCI repository, dataset contains one target variable and fifteen features(Age,Income,...)

Github Repository Link

<https://github.com/Lohilohith08/ML-based-credit-card-approval-system.git>

Models used:

MLModel Name	Accuracy	AUC	Precision	Recall	F1	MCC
Logistic Regression	0.8043	0.8816	0.8676	0.7662	0.8138	0.6146
Decision Tree	0.7899	0.7947	0.8529	0.7532	0.8000	0.5854
KNN	0.7971	0.8417	0.8101	0.8312	0.8205	0.5875
Naive Bayes	0.8261	0.8610	0.7912	0.9351	0.8571	0.6535
Random Forest	0.8333	0.9142	0.8553	0.8442	0.8497	0.6628

Observations:

ML Model Name	Observation about model performance
Logistic Regression	Model has an precision of 86.76%, indicating good performance in correctly identifying positive cases,However Recall is of 76.62% suggests that it misses some positive cases.Overall a strong model.
Decision Tree	Model's Accuracy and AUC score are not good, making it a weakest model in terms of overall predictive performance.
KNN	Models'Recall score is strong, means model has identifiedgood proportion of positive cases.However its precision is lower than Logistic Regression,Randon Forest.Overall KNN showed moderate performance.
Naive Bayes	Model has high Recall of 93.51% highest among all models, but Precision is comparatively lower suggesting model is highly efective in identifying positive case but also produces more false positives.
Random Forest	Model has produced best overall performance among all other models in terms of all scores.Therefore it appears to be the suitable model for the problem.